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Comparative effectiveness of exercise interventions for patellar tendinopathy: a systematic review and network meta-analysis of randomized controlled trials

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Abstract

Background Patellar tendinopathy (PT) is a common overuse condition in sport, but the comparative effectiveness of exercise-based rehabilitation strategies remains uncertain. Earlier syntheses often relied heavily on ranking metrics and included clinically heterogeneous intervention nodes. This study aimed to compare exercise interventions for PT while improving the transparency of intervention classification, transitivity assessment, and sensitivity analyses.

Methods This systematic review and network meta-analysis followed PRISMA-NMA guidance and was prospectively registered in PROSPERO (CRD420261296478). Eleven electronic databases were searched from inception to December 2025, and ClinicalTrials.gov was also screened for ongoing or unpublished trials. Randomized controlled trials evaluating exercise-based interventions for clinically and/or imaging-diagnosed patellar tendinopathy were eligible. The primary outcome was the Victorian Institute of Sport Assessment for Patellar Tendinopathy (VISA-P) score. Risk of bias was assessed using the Cochrane RoB 2.0 tool. Because the revised primary network was sparse and several comparisons were informed by limited evidence, the findings were interpreted cautiously. For the revised primary analysis, a conceptually cleaner connected network was constructed by excluding combined exercise-plus-adjunct interventions and one isolated 4-week comparison that did not connect to the main network. A random-effects frequentist network meta-analysis was conducted in R using the netmeta package, with heavy slow resistance training (HSR) as the reference treatment. Additional descriptive and sensitivity analyses were undertaken to examine effect-modifier distributions, follow-up window, and population heterogeneity.

Results Seventeen randomized controlled trials were included in the qualitative synthesis. The broader quantitative evidence framework comprised 14 studies (456 participants), which was retained as a sensitivity analysis. The revised primary connected network included 10 studies with 313 participants and 11 treatment nodes. Most trials were judged as having some concerns for overall risk of bias, particularly in domains related to deviations from intended interventions, where adherence, co-interventions, and the practical difficulty of blinding in exercise trials required careful consideration; no trial was judged as having overall high risk of bias. In the primary network, no intervention demonstrated statistical superiority over HSR, and the observed between-intervention differences did not indicate a

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clinically meaningful hierarchy. Eccentric overload showed the most favorable point estimate versus HSR (SMD 0.33, 95% CI -0.87 to 1.52), followed by blood flow restriction training (0.26, -0.40 to 0.92) and moderate slow resistance (0.12, -0.49 to 0.72). Flywheel training (-1.00, -1.65 to -0.35), eccentric step training (-1.30, -2.59 to -0.02), and concentric exercise (-2.59, -4.16 to -1.01) were estimated to perform less favorably than HSR. Probability-based rankings were exploratory and should not be interpreted as evidence of definitive or clinically meaningful treatment superiority.

Conclusions The revised evidence does not support clear superiority of any exercise intervention over HSR for short-term VISA-P improvement in PT, and the observed between-intervention differences do not indicate a clinically meaningful hierarchy. HSR remains a reasonable reference treatment, while several progressive loading-based strategies appear broadly comparable within the limits of the current evidence. Treatment rankings should not be interpreted as definitive, and future trials should use clearer intervention definitions, more comparable populations, and more consistent follow-up schedules.

Keywords Patellar tendinopathy, Exercise therapy, Network meta-analysis, Randomized controlled trials, Rehabilitation, Heavy slow resistance training

Introduction

Patellar tendinopathy (PT), commonly referred to as jumper's knee, is a frequent and often persistent overuse disorder in physically active populations [1, 2]. It is characterized by load-related anterior knee pain and localized tenderness at the inferior pole of the patella, and is particularly common in sports involving repetitive jumping, sprinting, and rapid changes of direction [3, 4]. The condition can substantially affect training continuity, competitive performance, and return-to-sport readiness, especially when symptoms become prolonged or recurrent.

Current understanding of PT is closely linked to tendon loading theory and to the continuum model of tendinopathy, in which tendon pathology is viewed as a dynamic response to mechanical load rather than a simple inflammatory process [5, 6]. Within this framework, exercise-based rehabilitation is regarded as the cornerstone of conservative management because it aims to restore tendon load tolerance, improve functional capacity, and address pain within a progressive loading context [7–9]. Compared with passive treatment approaches, structured exercise interventions are generally considered more relevant to long-term recovery [10].

Historically, isolated eccentric exercise was widely promoted as the standard exercise approach for PT [11, 12]. However, over time, concerns have emerged regarding its tolerability, adherence, and applicability across different patient groups, particularly when symptoms are highly irritable or when athletes remain exposed to substantial sport-specific loads [13, 14]. In parallel, alternative loading strategies have gained increasing attention. These include heavy slow resistance (HSR), progressive tendon-loading exercise (PTLE), blood flow restriction training (BFR), flywheel-based loading, and other exercise variants intended to improve pain and function through different mechanical or neuromuscular pathways [15–18].

As a result, the rehabilitation literature has expanded beyond a simple eccentric-versus-other comparison [19].

Despite this growth, the comparative effectiveness of available exercise options remains uncertain. Conventional pairwise meta-analyses are useful for direct comparisons, but they are limited when multiple active interventions coexist and head-to-head evidence is sparse [20]. Network meta-analysis offers the potential to integrate both direct and indirect evidence across several competing strategies [21]. However, the validity of such comparisons depends on assumptions that are particularly important in rehabilitation research, including conceptual coherence of intervention nodes, comparability of included populations, and reasonable alignment of follow-up time points [22]. These issues are especially relevant in PT, where studies vary in athletic level, symptom chronicity, baseline severity, and intervention content, and where some trials combine exercise with adjunctive modalities that may not be easily grouped into a single clinically coherent category [4, 23, 24].

Accordingly, the present systematic review and network meta-analysis sought to compare exercise-based interventions for PT using VISA-P as the primary outcome, while also addressing the methodological issues that complicate interpretation in this field. In addition to conducting a broader quantitative synthesis, we reconstructed a conceptually cleaner connected primary network, examined effect-modifier distributions relevant to transitivity, and undertook sensitivity analyses based on follow-up window and population characteristics. The aim was not simply to generate a treatment ranking, but to provide a more transparent and clinically interpretable comparison of current exercise-based rehabilitation strategies for PT.

Methods

Protocol and registration

This systematic review and network meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension statement for Network Meta-Analyses (PRISMA-NMA) [21]. The review protocol was prospectively registered in PROSPERO (CRD420261296478).

Search strategy and eligibility criteria

A comprehensive literature search was performed in 11 electronic databases, including PubMed, Embase, Web of Science, Cochrane Library, CINAHL, PsycINFO, Scopus, PEDro, Google Scholar, Wanfang Data, and CNKI, from database inception to December 2025. ClinicalTrials.gov was also searched for ongoing or unpublished trials. Search terms combined controlled vocabulary and free-text terms related to patellar tendinopathy, jumper's knee, exercise therapy, loading, and randomized controlled trials. The full search strategies for each database are provided in the Supplementary Material.

Studies were eligible if they met the following criteria. Participants had to have a clinical and/or imaging-supported diagnosis of patellar tendinopathy [4]. Eligible interventions included structured exercise-based rehabilitation programmes. Comparators included other exercise interventions, usual care, or non-exercise control conditions. The primary outcome was the Victorian Institute of Sport Assessment for Patellar Tendinopathy (VISA-P) score [25]. Only randomized controlled trials were included.

For the qualitative synthesis, all eligible exercise-based trials were retained. For quantitative synthesis, interventions were first categorized according to their dominant loading principle [19]. In the broader network, this included heavy slow resistance, progressive tendon-loading exercise, eccentric loading, blood flow restriction training, flywheel-based loading, and combined exercise-plus-adjunct strategies. However, because network validity depends on conceptual consistency of intervention nodes, the revised primary network was restricted to a connected set of studies with more coherent intervention definitions [22]. Studies involving combined exercise with adjunctive modalities were therefore not retained in the primary network and were instead considered in sensitivity analyses [23, 24]. In addition, one isolated comparison with a short 4-week follow-up that did not connect to the main network was excluded from the primary analysis. Studies combining exercise with injection therapies were excluded from all analyses in order to isolate the effects of mechanical loading interventions [18]. Outcome data contributing to the broader quantitative evidence framework are presented in Supplementary Table S6.

Data extraction and outcome selection

Data extraction was performed independently by two reviewers using a standardized form. Extracted information included study characteristics, participant characteristics, intervention details, comparator details, and outcome data. Particular attention was paid to variables considered potential effect modifiers, including age, symptom duration, baseline VISA-P score, athletic level, and follow-up duration.

For the primary quantitative analysis, final post-intervention VISA-P values were used whenever available. Mean and standard deviation were extracted preferentially. Where studies reported values in alternative formats, such as standard errors, medians with interquartile ranges, or graphical data only, these were handled transparently and considered separately when relevant to sensitivity analyses. The main analysis was based on the follow-up time point closest to 12 weeks after intervention initiation. This time point was selected a priori because it represented the most common short-term endpoint across the included trials, provided the greatest overlap of comparable post-intervention outcome data, and was consistent with the typical end-of-program assessment timing used in patellar tendinopathy exercise studies and prior syntheses [4, 18, 19]. To assess the influence of follow-up heterogeneity, an additional sensitivity analysis restricted to a narrower time window was planned. Study eligibility for this sensitivity analysis is summarized in Supplementary Table S2.

Risk of bias assessment

Risk of bias was assessed using the Cochrane Risk of Bias 2.0 tool [26]. This tool evaluates bias arising from the randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Given the nature of exercise trials, participant and therapist blinding was rarely feasible. Therefore, particular attention was given to deviations from intended interventions and to the handling of adherence, co-interventions, and protocol deviations [27]. Risk-of-bias figures were generated using robvis [28], and the resulting visual summaries are presented in Figs. 1 and 2.

Statistical analysis

A random-effects frequentist network meta-analysis was conducted in R using the netmeta package for the primary connected network [22]. Heavy slow resistance (HSR) was used as the reference treatment because it is widely regarded as a contemporary loading-based standard in patellar tendinopathy rehabilitation [7, 9, 15]. Comparative effects were expressed as standardized mean differences (SMDs) with 95% confidence intervals. Although VISA-P was used across studies [25], SMDs

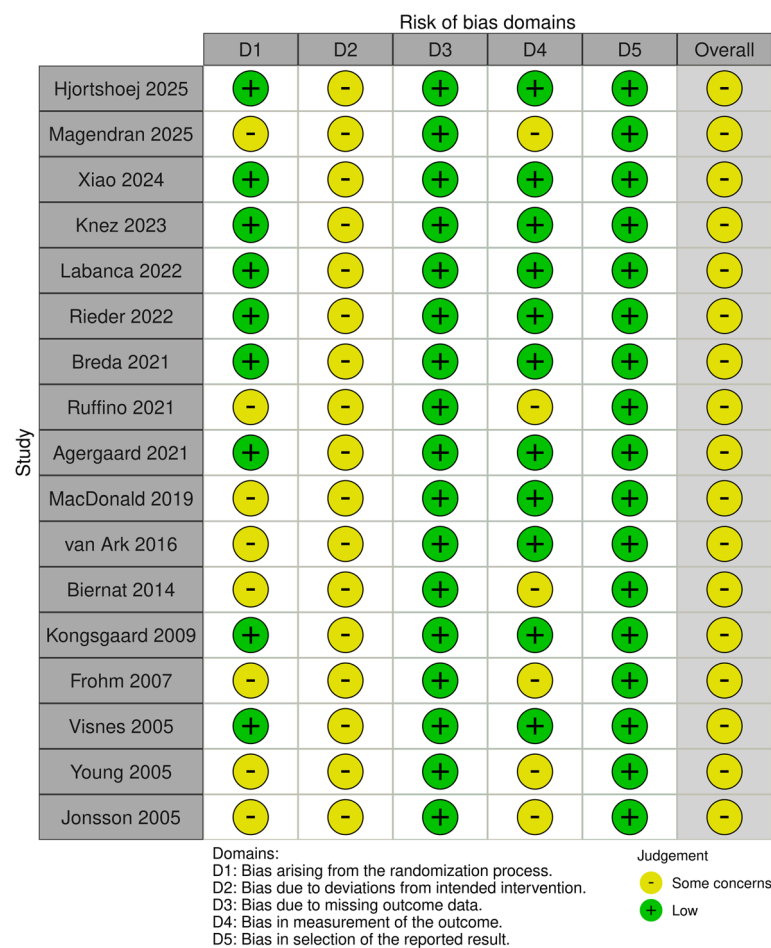


Fig. 1 Risk-of-bias traffic-light plot for all randomized controlled trials included in the qualitative synthesis (RoB 2.0). Legend: Traffic-light plot summarizing domain-level and overall RoB 2.0 judgments for all randomized controlled trials included in the qualitative synthesis. Note: Risk of bias was assessed across the following domains: bias arising from the randomization process; bias due to deviations from intended interventions; bias due to missing outcome data; bias in measurement of the outcome; and bias in selection of the reported result. These judgments apply to all 17 randomized controlled trials included in the qualitative synthesis and are not restricted to the revised primary network. Abbreviations: RoB 2.0, Cochrane Risk of Bias 2.0 tool

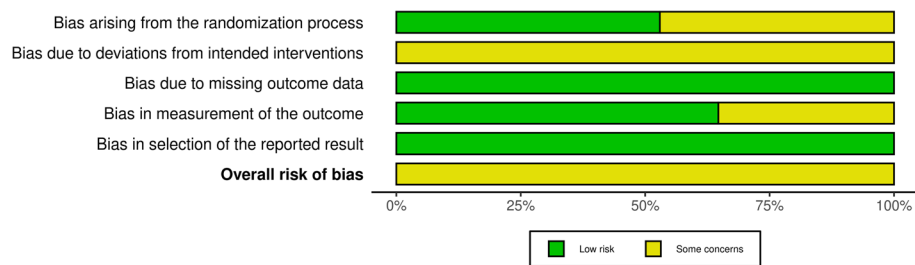


Fig. 2 Risk-of-bias summary plot for all randomized controlled trials included in the qualitative synthesis (RoB 2.0). Legend: Summary plot showing the proportion of studies judged at each RoB 2.0 level across domains and overall for all randomized controlled trials included in the qualitative synthesis. Note: This figure summarizes RoB 2.0 assessments for all 17 randomized controlled trials included in the qualitative synthesis. It is presented to characterize the overall bias profile of the evidence base and does not correspond only to the revised primary network or the broader sensitivity analysis. Abbreviations: RoB 2.0, Cochrane Risk of Bias 2.0 tool

were selected to preserve a consistent effect metric across comparisons because outcome dispersion and reporting formats varied across the quantitative evidence base. This approach also allowed more coherent comparison across the broader and sensitivity analyses, although it reduced direct clinical interpretability relative to mean differences. Network geometry was visualized to describe the structure of available direct comparisons, and the revised primary connected network is presented in Fig. 3.

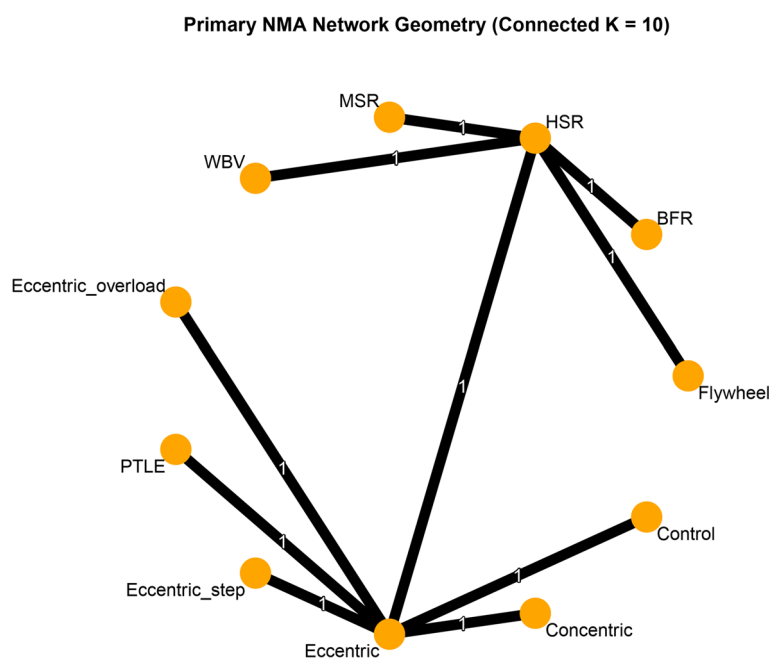


Fig. 3 Network plot for the revised primary analysis (connected K = 10 network). Legend: Nodes represent intervention categories included in the revised primary network, and connecting lines represent direct head-to-head comparisons between interventions. Node size is proportional to the total number of participants assigned to each intervention, and line thickness is proportional to the number of studies informing each direct comparison. Note: This figure presents the conceptually cleaner connected network used for the revised primary analysis. The revised primary network included 10 studies, 313 participants, and 11 treatment nodes. Combined exercise-plus-adjunct interventions and one isolated 4-week comparison that did not connect to the main network were excluded from this primary network and were considered only in sensitivity analyses. Abbreviations: HSR, heavy slow resistance; PTLE, progressive tendon-loading exercise; BFR, blood flow restriction training; WBV, whole-body vibration

The revised primary analysis was based on a connected K = 10 network. To improve node coherence, intervention labels were harmonized only where treatments reflected the same underlying clinical loading concept and were considered sufficiently similar in their exercise rationale and delivery. Closely related eccentric-loading protocols were grouped within the broader eccentric category when justified by intervention content, whereas distinct active variants were retained as separate nodes when collapsing them would have obscured clinically meaningful differences in loading strategy or programme structure. The original broader K = 14 network was retained as a sensitivity analysis to evaluate whether the main interpretation changed when a wider set of intervention categories was included.

Assessment of transitivity, heterogeneity, and inconsistency

Transitivity was assessed descriptively by comparing key effect modifiers across studies and comparisons [20], including age, symptom duration, baseline VISA-P score, athletic profile, and follow-up timing. These data were summarized in dedicated tables to improve transparency. Additional sensitivity analyses were planned by restricting the evidence base according to follow-up window, population characteristics, and overall risk-of-bias status

where applicable, in order to examine whether the main findings were robust to differences in timing, participant mix, and study quality. However, because no included study was judged as having overall high risk of bias under RoB 2.0, a sensitivity analysis excluding high-risk studies was not applicable.

Statistical heterogeneity and inconsistency were examined within the constraints of the available network. In the revised primary network, the number of studies per comparison was limited and few closed loops were available. As a result, formal heterogeneity and inconsistency statistics were interpreted conservatively [20, 29]. Global inconsistency was explored using a design-by-treatment approach, and local inconsistency was explored with node-splitting where feasible [20, 29]. Because the revised primary network was sparse, these analyses were considered supportive rather than definitive.

Ranking of interventions

Intervention hierarchy was explored using probability-based ranking metrics [30]. These rankings were treated as exploratory and were not interpreted in isolation. Particular emphasis was placed on the magnitude and precision of comparative effect estimates rather than on ranking alone. In line with this approach, ranking results were interpreted cautiously, particularly when

the network was sparse and treatment estimates were imprecise.

Results

Study selection

The database search identified 6,275 records. After removal of duplicates and screening, 17 randomized controlled trials met the eligibility criteria and were included in the qualitative synthesis [14, 15, 18, 31–44] (Tables 1, 2, and 3). In the original broader quantitative evidence framework, 14 studies involving 456 participants were included in the network meta-analysis. In the revised primary analysis, the network was reconstructed to improve conceptual consistency of intervention nodes and to retain a connected network structure. Three studies involving combined exercise-plus-adjunct strategies and one isolated comparison with a short 4-week follow-up were therefore not retained in the primary network. As a result, the revised primary NMA included 10 studies with 313 participants. The original broader K=14 network, which included combined-intervention nodes and a wider set of intervention categories, was retained as a sensitivity analysis to assess whether the main interpretation changed under a less restrictive network specification. Study eligibility for the 10–14 week time-window sensitivity analysis is presented in Supplementary Table S2. The study selection process is shown in Fig. 4.

Risk of bias

Risk of bias was assessed for all 17 trials using RoB 2.0 [26]. Overall, most studies were judged as having some concerns rather than low risk, and no included study was judged as having overall high risk of bias. This pattern was driven mainly by Domain 2, deviations from intended interventions. In exercise trials, this domain requires consideration of adherence, co-interventions, protocol deviations, and the practical difficulty of blinding [27]. By contrast, the randomization process and outcome measurement were generally less problematic. No major change to the overall risk-of-bias profile was introduced by restricting the primary network, because the revised K=10 analysis was derived from the same trial set evaluated in the qualitative review. These findings support cautious interpretation of comparative estimates, particularly where sample sizes were small or intervention adherence may have differed across studies. Because no included study met the RoB 2.0 threshold for overall high risk, a sensitivity analysis excluding high-risk studies was not performed. This pattern is particularly relevant in a sparse network, where even modest risks related to adherence or deviations from intended interventions may have a disproportionate influence on comparative estimates. The domain-level and overall RoB 2.0 judgments

are shown in Fig. 1, and the summary distribution of risk-of-bias judgments across domains is shown in Fig. 2.

Comparative effectiveness in the revised primary network

The revised K=10 network remained sparse, with HSR and eccentric loading representing the most connected nodes [15, 31, 36, 38]. The geometry of the revised primary connected network is shown in Fig. 3. As shown in Fig. 5, no intervention demonstrated statistical superiority over HSR for VISA-P [25]. However, several interventions were estimated to perform worse than HSR.

Relative to HSR, eccentric overload showed the most favorable point estimate (SMD 0.33, 95% CI –0.87 to 1.52) [33, 45], followed by blood flow restriction training (0.26, –0.40 to 0.92) [35, 46–48] and moderate slow resistance (0.12, –0.49 to 0.72) [31]. Control (SMD 0.09, 95% CI –0.98 to 1.17), whole-body vibration (SMD 0.06, 95% CI –0.74 to 0.87) [40], and progressive tendon-loading exercise (SMD 0.01, 95% CI –0.92 to 0.94) [18] also followed. Eccentric exercise was close to HSR (SMD –0.23, 95% CI –1.02 to 0.56) [11, 36]. By contrast, flywheel training (SMD –1.00, 95% CI –1.65 to –0.35) [33, 49], eccentric step training (SMD –1.30, 95% CI –2.59 to –0.02) [41], and concentric exercise (SMD –2.59, 95% CI –4.16 to –1.01) [37] were less favorable than HSR.

Taken together, these findings do not support clear superiority of any alternative intervention over HSR. Several loading-based interventions yielded point estimates close to the null relative to HSR, but the associated confidence intervals were wide. The overall pattern therefore suggests that several progressive loading-based interventions may yield broadly comparable short-term outcomes to HSR, but the available estimates are too imprecise to support strong claims of comparative superiority.

Ranking of interventions

Probability-based ranking was examined as an exploratory analysis only. In particular, differences in ranking should not be interpreted as evidence of statistically significant differences between interventions. Consistent with the reviewer concerns, ranking results were interpreted cautiously and were not considered sufficient evidence of treatment superiority in the absence of precise and robust between-node differences. Eccentric overload ranked highest (P-score 0.7733), followed by blood flow restriction training (0.7562), moderate slow resistance (0.6755), control (0.6637), whole-body vibration (0.6390), progressive tendon-loading exercise (0.6219), and HSR (0.5919). Eccentric exercise ranked lower (0.4405), followed by flywheel training (0.1840), eccentric step training (0.1430), and concentric exercise (0.0109). The ranking results from the connected broader sensitivity analysis excluding the isolated van Ark comparison [41] are presented in Supplementary Table S5.

Table 1 Characteristics of included randomized controlled trials in the qualitative synthesis (N = 17)

Study	Population (activity level)	Sample size, n (Exp/Con)	Age, years (Exp/Con)	Symptom duration, months (Exp/Con)	Intervention group (Exp)	Comparator group (Con)	Follow-up reported/ selected
Agergaard et al. (2021) [31]	Recreational athletes	21/21	28.8 ± 5.1/32.3 ± 4.9	6.9 ± 2.4/7.3 ± 2.9	HSR	MSR	12 weeks
Biernat et al. (2014) [42]	Junior volleyball players	15/13	17.7 ± 0.7/16.5 ± 0.8	NR	Eccentric (unstable surface)	Eccentric (stable surface)	12 weeks
Breda et al. (2021) [18]	Mixed athletes (recreational/competitive)	38/38	24 ± 4 (overall)	24 (12–60)	PTLE	Eccentric (decline board)	12 and 24 weeks; 12 weeks used in the revised primary network
Frohm et al. (2007) [39]	Athletes (competitive/recreational)	11/9	26 ± 8.4/28 ± 8.0	NR (< 3 months)	Eccentric overload	Eccentric (decline board)	12 weeks
Hjortshøj et al. (2025) [35]	Male athletes	16/20	30.0 ± 9.2/32.7 ± 10.8	10.0/9.5	LL-BFR	HSR	12 weeks
Jonsson et al. (2005) [37]	Athletes (soccer/runners)	10/5	25.7 ± 9.9/24.1 ± 6.4	15.4 ± 6.0/19.6 ± 20.3	Eccentric (decline board)	Concentric	12 weeks
Knez et al. (2023) [32]	Recreational athletes	35/35	25.6 ± 4.2 (overall)	3	Eccentric (17° decline board)	Eccentric (25° decline board)	12 weeks
Kongsgaard et al. (2009) [15]	Recreational athletes	13/12	31.7 ± 8.5/31.3 ± 8.3	18.8 ± 10.6/18.8 ± 13.0	HSR	Eccentric (decline board)	12 weeks
Labanca et al. (2022)† [44]	Physically active participants	14/14	31.4 ± 11.6/33.6 ± 10.2	NR (> 6 months)	Combined (NMES + exercise)	HSR	8 weeks
MacDonald et al. (2019)† [38]	Active-duty military personnel	14/17	26.9 ± 7.4/31.3 ± 5.6	15.3 ± 22.1/6.4 ± 3.3	Combined (eccentric + hip strengthening)	Eccentric (knee/decline-board based comparator)	12 weeks
Magendran et al. (2025) [43]	University volleyball players	15/15	21.1 ± 1.7/20.8 ± 1.4	NR	Aquatic plyometric training	Land plyometric training	6 weeks
Rieder et al. (2022)‡ [40]	Recreational athletes	13/11	25 ± 6/29 ± 6	41 ± 45/45 ± 58	WBV	HSR	12 weeks
Ruffino et al. (2021) [33]	Recreational athletes	20/21	27.5 ± 5.4/31.7 ± 8.7	9.5 ± 5.2/17.2 ± 16.4	Flywheel training	HSR	12 weeks
van Ark et al. (2016)‡ [41]	Volleyball/basketball players	10/10	22.9 ± 4.9/23.1 ± 4.7	30.8 ± 17.1/39.6 ± 39.1	Isometric exercise	Isotonic exercise	4 weeks
Visnes et al. (2005) [14]	Elite volleyball players	13/16	26.8 ± 4.6/26.4 ± 3.4	67 ± 44/79 ± 75	Eccentric (decline board)	Control (usual training/no additional training)	12 weeks

Table 1 (continued)

Study	Population (activity level)	Sample size, n (Exp/Con)	Age, years (Exp/Con)	Symptom duration, months (Exp/Con)	Intervention group (Exp)	Comparator group (Con)	Follow-up reported/selected
Xiao et al. (2024) [†] [34]	University athletes	17/17	21.4 ± 1.5/22.1 ± 1.6	7.2 ± 5.2/16.3 ± 18.7	Combined (HSR + therapeutic ultrasound)	HSR	8 weeks
Young et al. (2005) [36]	Elite volleyball players	9/8	27.3 ± 1.8 (overall)	NR	Eccentric (decline board)	Eccentric step training	12 weeks

Sample sizes in Table 1 are reported as presented in the original trial reports where available; outcome-specific sample sizes contributing to the revised primary network meta-analysis are presented in Table 2

Data are presented as mean ± standard deviation where available; when only overall values or non-parametric summaries were available, the original reporting format was retained

Symptom duration refers to the duration of symptoms before intervention

For multi-arm studies, only the exercise-based comparison relevant to the present review framework is shown

Abbreviations: HSR Heavy slow resistance, MSR Moderate slow resistance, PTLE Progressive tendon-loading exercise, LL-BFR Low-load blood flow restriction, NMES Neuromuscular electrical stimulation, WBV Whole-body vibration, NR Not reported

[†] Included in the qualitative synthesis but not retained in the revised primary connected network because the intervention involved a combined exercise-plus-adjunct strategy

^{*} Included in the qualitative synthesis and in the broader evidence framework, but not retained in the revised primary connected network

Table 2 Outcome data contributing to the revised primary connected network meta-analysis (K = 10)

Study	Year	Intervention group (Exp)	Comparator group (Con)	n (Exp)	n (Con)	Follow-up used	VISA-P, mean ± SD (Exp)	VISA-P, mean ± SD (Con)
Jonsson et al	2005	Eccentric exercise	Concentric exercise	10	5	12 weeks	83.3 ± 23.4	37.0 ± 4.6
Visnes et al	2005	Eccentric exercise	Control	13	16	12 weeks	70.2 ± 15.4	75.4 ± 16.7
Kongsgaard et al	2009	Heavy slow resistance (HSR)	Eccentric exercise	13	12	12 weeks	78.0 ± 18.0	75.0 ± 3.0
Breda et al	2021	Progressive tendon-loading exercise (PTLE)	Eccentric exercise	33	31	12 weeks	71.2 ± 13.8	67.7 ± 15.4
Agergaard et al	2021	Heavy slow resistance (HSR)	Moderate slow resistance (MSR)	21	21	12 weeks	70.5 ± 20.16	72.5 ± 13.29
Young et al	2005	Eccentric exercise	Eccentric step training	9	8	12 weeks	79.0 ± 10.0	68.0 ± 10.5
Rieder et al	2022	Whole-body vibration (WBV)	Heavy slow resistance (HSR)	13	11	12 weeks	84.0 ± 26.5	82.5 ± 18.6
Hjortshøj et al	2025	Blood flow restriction training (LL-BFR)	Heavy slow resistance (HSR)	16	20	12 weeks	77.0 ± 12.0	73.0 ± 17.9
Frohm et al	2007	Eccentric overload	Eccentric exercise	11	9	12 weeks	86.0 ± 15.6	75.0 ± 24.1
Ruffino et al	2021	Flywheel training	Heavy slow resistance (HSR)	20	21	12 weeks	72.0 ± 3.0	75.0 ± 3.0

Abbreviations: VISA-P Victorian Institute of Sport Assessment for Patellar Tendinopathy, SD Standard deviation, HSR Heavy slow resistance, PTLE Progressive tendon-loading exercise, LL-BFR Low-load blood flow restriction training, WBV Whole-body vibration, MSR Moderate slow resistance

a. This table summarizes the outcome data contributing to the revised primary connected network meta-analysis. The revised primary analysis included 10 studies, 313 participants, and 11 treatment nodes

b. For the primary quantitative analysis, the follow-up time point closest to 12 weeks after intervention initiation was used

c. Combined exercise-plus-adjunct interventions and one isolated 4-week comparison that did not connect to the main network were excluded from the revised primary network and were considered only in sensitivity analyses

d. This table is intended to document the primary connected network input set and therefore differs from the broader quantitative framework retained as a sensitivity analysis

Heterogeneity and inconsistency

Because the revised primary network was sparse and contained few closed loops, formal heterogeneity and inconsistency assessments were of limited informativeness [20, 29]. In the revised primary network, conventional heterogeneity statistics such as τ^2 and I^2 were not informative, and the design-by-treatment decomposition yielded Q

statistics of 0 with 0 degrees of freedom. Node-splitting did not identify evaluable direct–indirect contrasts for local inconsistency testing. These findings reflect the limited evaluability of inconsistency in the revised network rather than evidence of confirmed homogeneity or consistency. A sensitivity analysis excluding studies judged at overall high risk of bias was not performed because no

Table 3 Classification of studies across the qualitative synthesis, broader quantitative evidence framework, and revised primary connected network

Study	Included in qualitative synthesis (N = 17)	Included in broader quantitative evidence framework (K = 14)	Included in revised primary connected network meta-analysis (K = 10)	Classification note
Agergaard et al. (2021) [31]	Yes	Yes	Yes	Retained throughout; contributes the HSR versus MSR comparison
Biernat et al. (2014) [42]	Yes	No	No	Included in the qualitative synthesis only; not retained in the quantitative evidence framework
Breda et al. (2021) [18]	Yes	Yes	Yes	Retained throughout; contributes the PTLE versus eccentric exercise comparison
Frohm et al. (2007) [39]	Yes	Yes	Yes	Retained throughout; contributes the eccentric overload versus eccentric exercise comparison
Hjortshøj et al. (2025) [35]	Yes	Yes	Yes	Retained throughout; contributes the low-load blood flow restriction versus HSR comparison
Jonsson et al. (2005) [37]	Yes	Yes	Yes	Retained throughout; contributes the eccentric exercise versus concentric exercise comparison
Knez et al. (2023) [32]	Yes	No	No	Included in the qualitative synthesis only; not retained in the quantitative evidence framework
Kongsgaard et al. (2009) [15]	Yes	Yes	Yes	Retained throughout; contributes the HSR versus eccentric exercise comparison
Labanca et al. (2022) [44]	Yes	Yes	No	Retained in the broader quantitative evidence framework only; excluded from the revised primary network because it involved a combined exercise-plus-adjunct intervention
MacDonald et al. (2019) [38]	Yes	Yes	No	Retained in the broader quantitative evidence framework only; excluded from the revised primary network because it involved a combined exercise-plus-adjunct intervention
Magendran et al. (2025) [43]	Yes	No	No	Included in the qualitative synthesis only; not retained in the quantitative evidence framework
Rieder et al. (2022) [40]	Yes	Yes	Yes	Retained throughout; contributes the WBV versus HSR comparison
Ruffino et al. (2021) [33]	Yes	Yes	Yes	Retained throughout; contributes the flywheel versus HSR comparison
van Ark et al. (2016) [41]	Yes	Yes	No	Retained in the broader quantitative evidence framework only; excluded from the revised primary network because it was an isolated 4-week comparison that did not connect to the main network
Visnes et al. (2005) [14]	Yes	Yes	Yes	Retained throughout; contributes the eccentric exercise versus control comparison
Xiao et al. (2024) [34]	Yes	Yes	No	Retained in the broader quantitative evidence framework only; excluded from the revised primary network because it involved a combined exercise-plus-adjunct intervention
Young et al. (2005) [36]	Yes	Yes	Yes	Retained throughout; contributes the eccentric exercise versus eccentric step training comparison

Abbreviations: HSR Heavy slow resistance, MSR Moderate slow resistance, PTLE Progressive tendon-loading exercise, WBV Whole-body vibration

a. Seventeen randomized controlled trials were included in the qualitative synthesis. Fourteen studies were retained in the broader quantitative evidence framework as a sensitivity analysis framework, and 10 studies were retained in the revised primary connected network

b. The revised primary connected network excluded three studies involving combined exercise-plus-adjunct interventions and one isolated 4-week comparison (van Ark et al., 2016) that did not connect to the main network

c. Studies not retained in the broader quantitative evidence framework contributed to the qualitative synthesis only

included trial was judged as having overall high risk of bias under RoB 2.0. A separate sensitivity analysis excluding Rieder et al. [40], which used a waitlist-controlled design with later reallocation of participants, likewise did not materially alter the overall interpretation; the network remained connected, and the ranking pattern was broadly similar to that of the primary analysis. The

transitivity assumption was therefore further explored descriptively using effect-modifier tables and sensitivity analyses restricted by follow-up window and population characteristics.

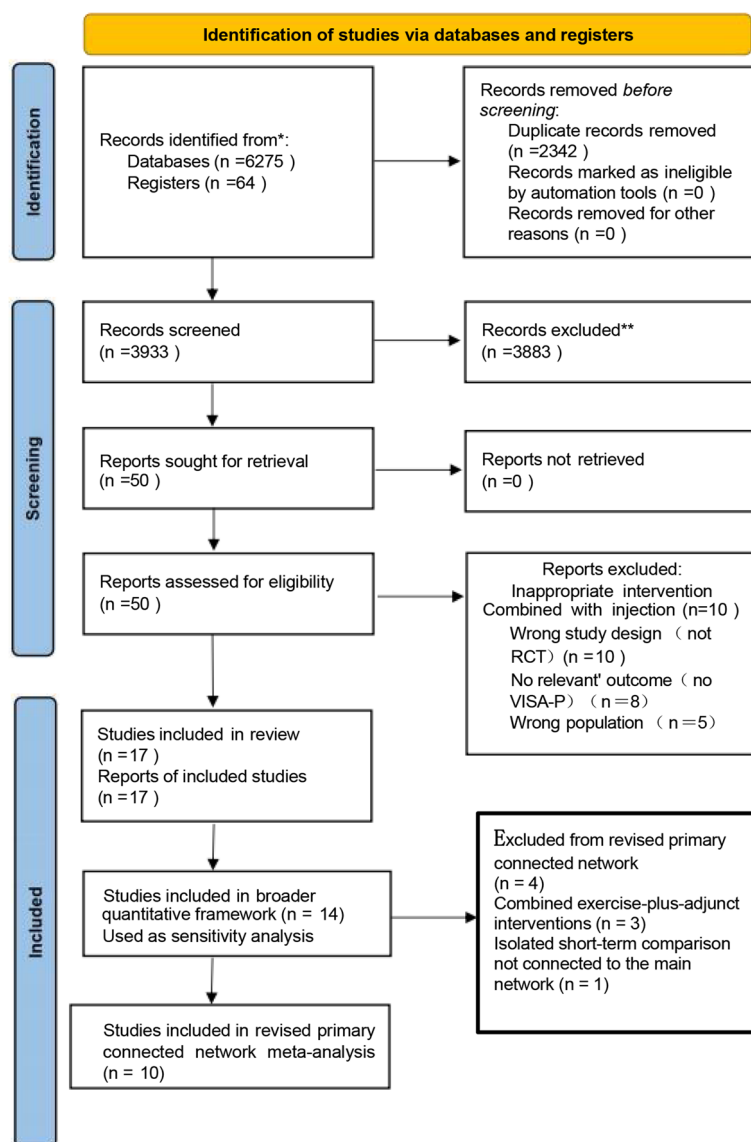


Fig. 4 PRISMA flow diagram for study selection. The database and register searches identified 6,275 records, of which 17 randomized controlled trials were included in the qualitative synthesis. Fourteen studies were retained in the broader quantitative evidence framework, which was considered in sensitivity analyses. The revised primary connected network meta-analysis included 10 studies after exclusion of combined exercise-plus-adjunct interventions and one isolated short-term comparison that did not connect to the main network. Abbreviations: PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Discussion

Principal findings

This revised network meta-analysis evaluated exercise-based interventions for patellar tendinopathy using a conceptually narrower and connected primary network. The main finding was that no intervention demonstrated statistical superiority over heavy slow resistance training (HSR), and the observed between-intervention differences were not suggestive of clinically important benefit relative to HSR [15, 31]. Several interventions yielded point estimates close to HSR, whereas concentric exercise, eccentric step training, and flywheel training were

estimated to perform less favorably. Overall, the revised network suggests that HSR remains a reasonable reference treatment, but that several loading-based alternatives may achieve broadly similar short-term outcomes. At the same time, the uncertainty around many estimates was substantial, and the results should be interpreted with caution.

Interpretation of the revised primary network

A key strength of the revised analysis is that the primary network was reconstructed to improve node coherence and preserve network connectivity. This was done in

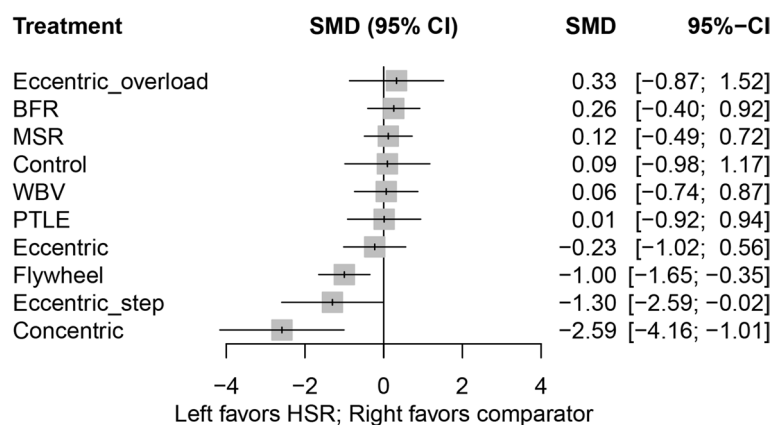


Fig. 5 Forest plot of comparative effects versus HSR in the revised primary network meta-analysis (connected K = 10 network). Legend: Forest plot showing random-effects standardized mean differences (SMDs) with 95% confidence intervals (CIs) for VISA-P in the revised primary network meta-analysis, using heavy slow resistance (HSR) as the reference treatment. Positive values indicate more favorable estimates relative to HSR, whereas negative values indicate less favorable estimates relative to HSR. Values with confidence intervals crossing 0 indicate no clear difference from HSR. Note: The revised primary analysis included 10 studies, 313 participants, and 11 treatment nodes. No intervention demonstrated statistical superiority over HSR in the revised primary network. Because the network was sparse and several estimates were imprecise, these comparative results should be interpreted cautiously. Abbreviations: HSR, heavy slow resistance; SMD, standardized mean difference; CI, confidence interval; VISA-P, Victorian Institute of Sport Assessment for Patellar Tendinopathy; BFR, blood flow restriction training; PTLE, progressive tendon-loading exercise; WBV, whole-body vibration

response to concerns that the broader network may have combined interventions with different mechanisms into overly broad categories. In particular, combined exercise-plus-adjunct interventions were not retained in the primary network because their inclusion raised concerns about conceptual heterogeneity and interpretability. The broader K = 14 network was retained as a sensitivity analysis, allowing the robustness of the overall interpretation to be examined without relying on that node in the primary model.

Within the revised K = 10 network, several active loading-based strategies showed estimates close to HSR, including blood flow restriction training [46, 47], moderate slow resistance [31], whole-body vibration [40], and progressive tendon-loading exercise [18]. These findings do not establish equivalence or non-inferiority, but they suggest that the currently available evidence does not clearly favor one progressive loading approach over another. By contrast, concentric exercise [37] ranked lowest and was associated with less favorable outcomes than HSR, which is consistent with earlier concerns that isolated concentric loading may not reflect the clinical demands of patellar tendon rehabilitation as well as progressive loading-based programmes.

Ranking results and clinical interpretation

The ranking analysis should be interpreted cautiously. Although eccentric overload, blood flow restriction training, and moderate slow resistance appeared near the top of the P-score hierarchy, the network was sparse and many comparisons were imprecise. For that reason, the ranking results were considered exploratory rather than confirmatory. This point is important because

probability-based ranking can appear more definitive than the underlying estimates justify, particularly in small networks with limited direct evidence.

The clinical relevance of between-intervention differences also requires restraint in interpretation. In the original broader network, combined exercise-plus-adjunct interventions were considered only in a supplementary sensitivity analysis because that broader specification was clinically more heterogeneous and less directly interpretable than the revised primary network. In the revised primary network, no intervention showed clear superiority over HSR. The magnitude of these differences should not be assumed to represent clinically important benefit, particularly given the sparse network and the imprecision of many estimates. Where broader-network estimates suggested an apparent hierarchy, the magnitude of between-intervention differences was modest and should not be interpreted as evidence of clinically meaningful superiority. Taken together, these results suggest that treatment selection should not be based on ranking alone. Rather, the current evidence supports the view that several structured loading programmes may be reasonable options, while the choice of intervention may need to reflect symptom tolerance, feasibility, supervision, and patient preference. Accordingly, ranking results should be viewed mainly as exploratory signals rather than as a basis for changing clinical practice.

Transitivity and population heterogeneity

The validity of indirect comparisons in network meta-analysis depends on the plausibility of transitivity. In the present review, this assumption was explored descriptively by summarizing key effect modifiers, including

age, symptom duration, baseline VISA-P score, athletic profile, and follow-up duration. Although no obvious descriptive imbalance was identified across the main comparisons, the available information was limited and the included populations were not fully homogeneous. Some trials enrolled elite volleyball players [14, 36], others involved recreationally active participants [31, 33, 40], and one study included active-duty military personnel [38]. These differences may influence treatment response through variation in training load, exposure, and functional demands [50, 51].

To improve transparency, effect-modifier distributions were summarized in dedicated tables, and additional sensitivity analyses were undertaken by follow-up window and population restrictions. These analyses were intended to test whether the main interpretation was materially altered when more comparable subsets of studies were considered. This approach does not eliminate uncertainty, but it strengthens the interpretability of the revised primary analysis. Although longer-term outcomes remain important in patellar tendinopathy, short-term outcomes remain clinically relevant because they inform early rehabilitation decision-making, short-cycle programme evaluation, and the adjustment of loading strategies during the initial phase of care.

Limitations

This review has several limitations. First, the primary network remained small and sparse, with relatively few studies informing each comparison. As a result, many confidence intervals were wide, and formal heterogeneity and inconsistency metrics were of limited value. Second, some outcome data in the broader evidence base were available only in non-standard formats, including medians, standard errors, or graph-derived estimates, which limited the extent to which all studies could contribute equally to the primary model. Third, the broader $K=14$ network contained clinically heterogeneous combined-intervention nodes. We addressed this by excluding those nodes from the primary analysis and retaining the broader network as a sensitivity analysis; however, adjunct-specific subgroup network meta-analysis was not feasible because too few studies were available within each adjunct category. Fourth, one included study (Rieder et al. [40]) used a waitlist-controlled design with later reallocation of participants. Although the initial randomized WBV versus HSR comparison was retained in the primary network, this design feature should still be considered when interpreting the corresponding network estimate. Finally, risk of bias [26] was commonly rated as having some concerns, particularly in relation to deviations from intended interventions, adherence, and the practical impossibility of blinding in exercise trials [27]. No included study was judged as having overall high risk

of bias, which limited the feasibility of a sensitivity analysis based on exclusion of high-risk studies. These issues may have biased estimates in either direction depending on differential adherence, co-interventions, or selective behavior changes across groups, and they therefore add uncertainty rather than supporting a consistent directional interpretation of treatment effects.

Implications

Overall, the revised findings do not indicate clear superiority of any intervention over HSR. HSR remains a reasonable reference treatment, while several alternative loading-based strategies appear broadly comparable within the limits of the current evidence [15, 31]. The main message is therefore not that one exercise approach is definitively best, but that progressive loading remains central [5, 7, 9] and that current comparative evidence is still limited by sparse networks, imprecision, and clinical heterogeneity [20, 22]. Larger, better-aligned trials with clearer intervention definitions and more consistent follow-up schedules are needed to strengthen future comparative inferences [10, 52]. Importantly, the absence of statistically significant superiority over HSR should not be interpreted as evidence of equivalence or non-inferiority between interventions.

Conclusion

In this revised network meta-analysis, no intervention demonstrated clear statistical superiority over HSR in the primary connected network, and the observed between-intervention differences were unlikely to represent clinically important benefit. Although some interventions ranked above HSR probabilistically, these rankings should be interpreted cautiously because the network was sparse and several estimates were imprecise. The current evidence suggests that HSR remains a reasonable reference treatment [15, 31], while several alternative loading-based interventions may achieve broadly comparable short-term outcomes. These findings do not support strong claims of treatment hierarchy. However, the absence of clear superiority should not be interpreted as formal evidence of equivalence or non-inferiority. Instead, intervention choice should be guided by clinical context, symptom tolerance, feasibility, and availability of supervision [7, 9, 53]. Future trials should use clearer intervention definitions, more comparable populations, and more consistent follow-up schedules [10, 52].

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13102-026-01743-4>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.
 Supplementary Material 4.
 Supplementary Material 5.
 Supplementary Material 6.
 Supplementary Material 7.
 Supplementary Material 8.
 Supplementary Material 9.
 Supplementary Material 10.
 Supplementary Material 11

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Authors' contributions

YL and CL conceived the study design and performed the literature search. YL and FY conducted data extraction and risk of bias assessment. YL performed the statistical analysis. YL drafted the manuscript. FY supervised the study and revised the manuscript critically for important intellectual content. All authors read and approved the final manuscript.

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Data availability

The extracted outcome data supporting the findings of this study are provided in the Supplementary Material. Additional analytic materials are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This study is a systematic review and meta-analysis of previously published data.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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